



School of Advanced Technology
Final Year Project
Project Specification Report

Project Title: Improving U-Net-Based Segmentation of Overlapping Organoids Using Boundary Supervision and Distance Map Regression

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Project field: Computer Vision / Biomedical Image Analysis / Deep Learning

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1- Project Description and Problem Statement

Organoids are three-dimensional biological models that can simulate the structure and function of real organs. They are widely used in disease modelling, drug screening and personalised medicine research. Through microscopic image analysis, researchers can obtain quantitative information such as the number, area, shape, boundary and growth status of organoids. Therefore, accurate image segmentation is a key step in organoid image analysis.

However, organoid microscopic image segmentation remains challenging. Organoids often have irregular shapes, large size variation, blurred boundaries and low image contrast. In addition, during culture and imaging, multiple organoids may touch, adhere to each other or overlap. In these cases, ordinary segmentation models may incorrectly merge several organoids into one connected region, leading to under-segmentation. Previous work on overlapping organoid segmentation also indicates that overlap regions usually account for only a small proportion of the training data, which makes them easy for models to ignore [3].

U-Net is a widely used deep learning model for medical image segmentation. Its encoder-decoder structure extracts semantic features from images, while skip connections help recover spatial details. This makes U-Net a suitable baseline for biomedical semantic segmentation [1], [2]. Nevertheless, a basic U-Net usually performs binary semantic segmentation, which means that it mainly distinguishes foreground objects from the background. When organoids touch or overlap, a simple binary mask cannot fully describe the boundary relationships between different organoid instances.

Therefore, this project proposes to improve a basic U-Net by introducing boundary supervision and distance map regression. The model will learn not only the overall organoid region, but also the contact boundaries and instance-related spatial information. Related studies suggest that boundary cues and distance maps can improve the segmentation of overlapping or touching objects [4]. Since boundary and overlap regions usually occupy only a small number of pixels, this project will also consider loss functions such as Focal Loss to reduce the impact of class imbalance [5]. Finally, watershed post-processing will be used to further separate connected organoid regions into individual instances, with the aim of improving segmentation performance on touching or overlapping organoids.

2- Aims and Objectives

The overall aim of this project is to design and evaluate an improved U-Net-based segmentation method for overlapping or touching organoids in microscopic images. The proposed method will extend the basic U-Net by adding boundary prediction and distance map regression, and will combine these predictions with post-processing to achieve improved instance separation.

- review literature on U-Net-based medical image segmentation, organoid image analysis, and segmentation of touching or overlapping objects;
- implement a basic U-Net model for binary segmentation of organoid regions and use it as the baseline model;
- inspect the available organoid annotations and determine whether the labels are binary masks or instance-level masks;
- generate additional supervisory labels, such as boundary maps and distance maps, based on the available annotations;
- design a multi-task U-Net that can predict organoid masks, boundary maps and distance maps at the same time;
- apply suitable loss functions, such as Dice Loss, Focal Loss and regression loss, to address class imbalance between background, organoid regions and boundary regions;
- use watershed post-processing to split connected or touching organoid regions into separate instances;
- evaluate the baseline and improved models using pixel-level metrics such as Dice, IoU and F1-score, as well as instance-level metrics such as AJI and object-level F1;
- analyse whether boundary supervision, distance map regression and watershed post-processing can effectively improve the segmentation of overlapping organoids.

3- Project Plan: Tasks and Milestones (Gantt Chart)

Week	Main Focus	Stage Content	Expected Milestone
Week 1	Background study and baseline understanding	U-Net, medical image segmentation and organoid image analysis are reviewed.	Research background and baseline model understanding are established.
Week 2	Overlapping organoid problem	Touching-object	Project specification and

	and research design	segmentation, boundary-aware segmentation and distance map methods are reviewed.	overall research route are completed.
Week 3	Dataset and annotation preparation	Microscopic images and masks are organised. Annotation type and dataset split are confirmed.	Dataset structure and annotation type are confirmed.
Week 4	Basic U-Net baseline	A basic U-Net is implemented and trained for binary organoid segmentation.	Baseline results and initial evaluation metrics are obtained.
Week 5	Auxiliary label generation	Boundary maps and distance maps are generated from the available annotations.	Auxiliary labels are ready for multi-task learning.
Week 6	Improved multi-task U-Net	A multi-task U-Net is trained to predict masks, boundaries and distance maps.	Improved model results are obtained and compared with the baseline.
Week 7	Watershed post-processing and instance separation	Watershed post-processing is applied to separate touching organoid regions.	Instance separation results and instance-level metrics are obtained.
Week 8	Result analysis and final preparation	Results are analysed and final report, presentation slides and code are prepared.	Final report, presentation materials and code package are completed.

4- Project Deliverables

By the end of the project, several deliverables are expected to be completed for the final report, presentation and model demonstration.

1. Literature review: A concise review of U-Net-based medical image segmentation, organoid image analysis and overlapping-object segmentation will be produced. This review will provide the theoretical basis for the proposed method.

2. Prepared dataset: The dataset will include organised raw organoid microscopic images, segmentation annotations, training/validation/test splits and, where possible, generated boundary maps and distance maps.

3. Basic U-Net baseline model: A basic U-Net model will be implemented and trained for binary organoid mask segmentation. Its results will be used as the baseline for comparison.

4. Improved multi-task U-Net model: An improved model will be developed to predict organoid masks, boundary maps and distance maps simultaneously. This model is expected to improve the representation of touching and overlapping regions.

5. Watershed post-processing workflow: A watershed-based post-processing workflow will be used to separate connected or touching organoid regions into individual instances.

6. Quantitative evaluation results: The baseline and improved models will be evaluated using metrics such as Dice, IoU, F1-score, AJI and object-level F1.

7. Visual comparison figures: The final report will include visual comparisons among the original image, ground-truth annotation, baseline U-Net prediction, improved model prediction and instance separation result.

8. Final poster & code: The final project poster and presentation slides and code will be prepared as the final project outputs.

5- Project Industrial Relevance

This project is closely related to biomedical image analysis, drug discovery, precision medicine and high-throughput experimental platforms. Organoids are important in vitro biological models and have been widely used for disease modelling, drug response testing and personalised treatment evaluation. In these applications, researchers often need to analyse large numbers of microscopic organoid images, including counting organoids, measuring area and morphology, monitoring growth trends and assessing responses to drug treatment.

Traditional manual segmentation is time-consuming, subjective and difficult to scale to large image datasets. Deep learning-based automatic segmentation can significantly improve analysis efficiency and reduce the cost of manual annotation and measurement. However, overlapping or touching organoids remain a major difficulty. If a model cannot separate adjacent organoids correctly, it may lead to inaccurate counting, incorrect area measurement and biased downstream analysis.

Recent work on overlapping organoid instance segmentation further suggests that this task requires not only detecting overlap regions, but also recovering each independent organoid instance as accurately as possible. Some newer studies have started to use pseudo-label correction, mask decomposition and synthesis-assisted learning to improve overlapping organoid instance segmentation [6]. Therefore, by improving the ability of U-Net to segment overlapping organoids, this project can contribute to more accurate and automated organoid image analysis. In the future, similar methods could be applied in biomedical research laboratories, organoid-based drug screening platforms, medical image analysis software and high-throughput image processing workflows in pharmaceutical research.

References

- [1] N. Siddique, S. Paheding, C. P. Elkin, and V. Devabhaktuni, "U-Net and its variants for medical image segmentation: A review of theory and applications," *IEEE Access*, vol. 9, pp. 82031-82057, 2021.
- [2] T. Zhou, Y. Dong, B. Huo, S. Liu, and Z. Ma, "U-Net and its applications in medical image segmentation: a review," *Journal of Image and Graphics*, vol. 26, no. 9, pp. 2058-2077, 2021.
- [3] A. Haja, S. Radu, and L. Schomaker, "A comparison of different U-Net models for segmentation of overlapping organoids," in *Proc. 9th International Conference on Biomedical and Bioinformatics Engineering (ICBBE)*, Kyoto, Japan, 2022, pp. 1-10.
- [4] X. Li, Y. Wang, Q. Tang, Z. Fan, and J. Yu, "Dual U-Net for the segmentation of overlapping glioma nuclei," *IEEE Access*, vol. 7, pp. 84040-84052, 2019.
- [5] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal Loss for Dense Object Detection," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017, pp. 2980-2988.

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- [6] G. Huang et al., "Boosting overlapping organoid instance segmentation using pseudo-label unmixing and synthesis-assisted learning," arXiv preprint arXiv:2601.06642, 2026.